BIO 005 Human Physiology · Week 8 notes · Dr. Sharilyn Rennie

The hypothalamus and pituitary

The written notes for the the hypothalamus and pituitary walkthrough. Week 8 competencies 12 to 15.

To print: these notes as a two-column PDF. Already printed the Week 8 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: where the hormones travel

In your textbook: Silverthorn Chapter 8, the parts on the hypothalamus and pituitary gland and on feedback loops.

ADH, from a desert hike to the kidney, step by step

1. You sweat and do not drink, and your plasma osmolarity climbs above the normal of about 290 mOsm/L.
2. Osmoreceptors in the hypothalamus lose a little water, shrink, and fire faster.
3. They excite the ADH neurons, whose cell bodies are in the hypothalamus. ADH was made there and packed into vesicles, which traveled down the axons through the stalk and are stored in the axon terminals in the posterior lobe.
4. Action potentials run down the axons. The terminals release ADH by exocytosis into capillaries in the posterior lobe, and it enters the general circulation.
5. ADH reaches the kidneys and binds V2 receptors on the blood side of the collecting duct cells.
6. Through cAMP, vesicles holding aquaporin-2 water channels move to the urine side of the cell and fuse with the membrane.
7. Water moves by osmosis out of the urine, through the cell, and back into the blood. You make a small amount of dark, concentrated urine.
8. The same rise in osmolarity also makes you thirsty.

Oxytocin in labor, step by step

1. The baby's head presses on the cervix and stretches it.
2. Sensory signals travel to the oxytocin neurons in the hypothalamus.
3. The neurons fire, and oxytocin stored in their terminals in the posterior lobe is released into the blood.
4. Oxytocin makes the smooth muscle of the uterus contract.
5. The contraction pushes the head harder against the cervix, which releases still more oxytocin: positive feedback, until the baby is born.

The cortisol axis, step by step

1. Tier one: neurons in the hypothalamus release corticotropin-releasing hormone (CRH) onto the first capillary bed at the top of the stalk.
2. CRH travels down the portal vessels in the stalk to the second capillary bed in the anterior lobe, still at high concentration.
3. Tier two: CRH makes anterior lobe cells release adrenocorticotropic hormone (ACTH) into the lobe's veins and the general circulation.
4. ACTH reaches the adrenal cortex, the outer layer of the adrenal gland on top of each kidney.
5. Tier three: the adrenal cortex releases cortisol into the general circulation.
6. Long loop: cortisol acts on the hypothalamus and the anterior lobe and lowers CRH and ACTH release.
7. Short loop: ACTH acts on the hypothalamus and lowers CRH release.

The thyroid axis, step by step

1. The hypothalamus releases thyrotropin-releasing hormone (TRH) into the portal vessels.
2. TRH makes the anterior lobe release thyroid-stimulating hormone (TSH) into the general circulation.
3. TSH makes the thyroid gland release thyroid hormone, mostly T4, which tissues convert to the more active T3.
4. Thyroid hormone feeds back on the hypothalamus and the anterior lobe and lowers TRH and TSH release.

Two glands in one

In your textbook: Silverthorn Chapter 8, the part on the hypothalamus and pituitary gland.

The hypothalamus is a small region in the floor of the brain that keeps internal conditions steady: water balance, body temperature, hunger and thirst, and the release of many hormones. The pituitary gland hangs from it on a thin stalk and sits in a small cup of bone in the base of the skull.

	Posterior pituitary	Anterior pituitary
Also called	Neurohypophysis	Adenohypophysis
Tissue	Neural tissue: an extension of the hypothalamus	True endocrine tissue: epithelial gland cells
Where it forms before birth	Grows down from the brain	Grows up from the roof of the embryo's mouth
Its hormones	ADH (vasopressin) and oxytocin, made in hypothalamic cell bodies; the lobe stores and releases them	TSH, ACTH, GH, prolactin, FSH and LH, made by its own cells
How the hypothalamus controls it	Directly, through axons that run down the stalk	Through releasing and inhibiting hormones carried in the portal vessels

The posterior pituitary

In your textbook: Silverthorn Chapter 8, the part on the posterior pituitary.

Neurohormone

A hormone released into the blood by a neuron. ADH and oxytocin are neurohormones.

Antidiuretic hormone (ADH), or vasopressin

A peptide nine amino acids long. Main stimulus: rising plasma osmolarity. Also released when blood pressure or blood volume drops a lot, as after heavy bleeding, sensed by stretch receptors in the heart and blood vessels. Target: the collecting ducts of the kidneys, where it keeps water.

Oxytocin

A peptide nine amino acids long. Stimulus: stretch of the cervix in labor, and suckling at the breast. Targets: the smooth muscle of the uterus, and the cells around the milk glands.

Osmoreceptor

A neuron in the hypothalamus that senses plasma osmolarity. It shrinks a little and fires faster when the plasma gets more concentrated.

Aquaporin-2

The water channel that ADH puts into the urine side of collecting duct cells.

- ADH and oxytocin are made in the cell bodies of two separate groups of hypothalamic neurons, packed into vesicles, carried down the axons through the stalk, and stored in the axon terminals in the posterior lobe. When the neuron fires, they are released by exocytosis into capillaries and enter the general circulation.
- Normal plasma osmolarity is about 290 mOsm/L. Above about 280 mOsm/L, ADH release rises as osmolarity climbs. Rising osmolarity also causes thirst.
- ADH binds the V2 receptor on the blood side of a collecting duct cell. Through cAMP, vesicles carrying aquaporin-2 fuse with the urine side membrane, so water leaves the urine by osmosis and returns to the blood. The blood side of the cell always has water channels.
- High ADH: a small amount of dark, concentrated urine. Low ADH: the channels are pulled back into the cell, and a lot of pale, dilute urine.
- Diabetes insipidus: too little ADH, or kidneys that cannot respond to it. Large amounts of dilute urine and constant thirst.
- Oxytocin in labor contracts the uterus by positive feedback. A synthetic form is given to start or strengthen labor.
- Oxytocin during breastfeeding causes milk ejection (let-down). Making the milk is prolactin's job, from the anterior lobe.
- Both hormones leave the same lobe, but each comes from its own neurons, so the stimulus decides which one is released.

The anterior pituitary

In your textbook: Silverthorn Chapter 8, the parts on the anterior pituitary and the hypothalamic-hypophyseal portal system.

- Neurons in the hypothalamus release releasing hormones and inhibiting hormones onto a first capillary bed at the top of the stalk.
- Portal vessels run down the stalk to a second capillary bed in the anterior lobe. A portal system is any set of vessels that links two capillary beds directly; this one is the hypothalamic-hypophyseal portal system.
- The hypothalamus releases only tiny amounts of these hormones. The portal system delivers them straight to the anterior lobe while their concentration is still high, instead of diluting them in about five liters of blood.

Trophic hormone

A hormone that controls the release of another hormone, usually by another endocrine gland. Most anterior lobe hormones are trophic.

Anterior lobe hormone	Main target	Main control from the hypothalamus
Thyroid-stimulating hormone (TSH)	Thyroid gland	Thyrotropin-releasing hormone (TRH)
Adrenocorticotropic hormone (ACTH)	Adrenal cortex	Corticotropin-releasing hormone (CRH)
Growth hormone (GH)	Liver (which answers by making growth factors) and most other tissues	A releasing hormone, and an inhibiting hormone, somatostatin
Prolactin	Breast: milk production	Held back mainly by dopamine; when dopamine falls, prolactin rises
Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)	Ovaries or testes	Gonadotropin-releasing hormone (covered in the next walkthrough)

All six anterior lobe hormones are peptides or proteins.

Three-tier axes and feedback

In your textbook: Silverthorn Chapter 8, the part on feedback loops and the hypothalamic-pituitary pathways.

A three-tier axis is a chain of three hormones: a hypothalamic releasing hormone, an anterior pituitary trophic hormone, and the hormone of a third gland. Only the first one travels in the portal vessels; the other two travel in the general circulation.

Tier	Cortisol axis	Thyroid axis
1. Hypothalamus	CRH	TRH
2. Anterior pituitary	ACTH	TSH
3. Last gland	Adrenal cortex: cortisol	Thyroid gland: T4 and T3
What the last hormone does	Helps the body cope with stress, partly by raising blood glucose	Sets how fast many tissues use energy

Long-loop negative feedback

The last gland's hormone acts on the hypothalamus and the anterior pituitary and lowers their release. It is the main feedback signal in these axes.

Short-loop negative feedback

The pituitary hormone acts on the hypothalamus and lowers releasing hormone release.

- ACTH also keeps the adrenal cortex healthy and full size.
- The thyroid releases mostly T4, and tissues convert it to the more active T3.
- If thyroid hormone rises above normal, TRH and TSH fall. If it falls, TRH and TSH rise.

Primary and secondary disorders

In your textbook: Silverthorn Chapter 8, the part on endocrine pathologies.

Hypersecretion

Too much hormone.

Hyposecretion

Too little hormone.

Primary disorder

The fault is in the last gland of the axis. Through feedback, the trophic hormone moves in the opposite direction to the last gland's hormone.

Secondary disorder

The fault is in the anterior pituitary. The trophic hormone and the last gland's hormone move in the same direction.

Disorder	Cortisol	ACTH	Example
Primary hyposecretion	Low	High	Addison disease (primary adrenal insufficiency): the adrenal cortex is damaged, most often by the immune system; high ACTH can darken the skin
Secondary hyposecretion	Low	Low	Secondary adrenal insufficiency: damage to the ACTH-making cells of the anterior lobe
Primary hypersecretion	High	Low	A tumor of the adrenal cortex that makes cortisol on its own
Secondary hypersecretion	High	High	Cushing disease: a pituitary tumor that makes too much ACTH and ignores feedback

- Cushing syndrome is the picture caused by high cortisol from any cause: high blood glucose, muscle protein breakdown, thin skin, and fat gathered in the face and trunk.
- One lab value, the trophic hormone, separates a primary from a secondary disorder when the last gland's hormone is low in both.
- The same rule works for the thyroid: low thyroid hormone with high TSH is primary hypothyroidism; low thyroid hormone with low TSH is secondary hypothyroidism.

Six months of prednisone, step by step

1. Prednisone acts like cortisol. It is taken in large daily doses for conditions such as severe asthma or rheumatoid arthritis.
2. It feeds back on the hypothalamus and the anterior lobe just as cortisol would, so CRH and ACTH fall very low.
3. Without ACTH, the adrenal cortex makes little cortisol of its own and shrinks over the months.
4. If the drug is stopped suddenly, the axis cannot restart right away, and the shrunken cortex cannot make enough cortisol: adrenal insufficiency, which can be life threatening, especially under stress.
5. So long courses are tapered, lowering the dose slowly while the hypothalamus, pituitary and adrenal cortex recover.

A cut pituitary stalk

- Every anterior lobe hormone except prolactin falls, because the releasing hormones can no longer reach the lobe.
- Prolactin rises, because the dopamine that holds it back is cut off too.
- ADH and oxytocin release from the posterior lobe stops, because the axons that bring them down are cut.

Measuring a hormone

In your textbook: Silverthorn Chapter 8, the parts on endocrine pathologies and on diagnosing endocrine problems.

1. An antibody that grabs only the hormone you want is stuck to the bottom of each well.
2. The sample is added, and the hormone in it is captured.
3. A second antibody carrying an enzyme sticks to the captured hormone.
4. The enzyme turns a colorless chemical colored. The more hormone in the well, the stronger the signal.
5. Wells with known amounts of hormone, the standards, are plotted as signal against amount to make a standard curve.
6. A patient's signal is read across to the curve and down to the amount axis.

- Worked example: a tired, cold patient who is gaining weight has low thyroid hormone. Her TSH reads about 12 mU/L. A typical adult reference range is roughly 0.4 to 4 mU/L, and each lab sets its own, so hers is high: primary hypothyroidism.
- A signal above the highest standard: dilute the sample and run it again, because the curve may not keep going straight past the standards.
- An antibody that also grabs a similar molecule adds that molecule to the signal, so the number is only as good as the antibody.